

Resit exam (1^h30)

Name :

Groupe:

Exercice 1. We took note of the performance bonuses (in DA) of the employees of a company. The results are given in the following statistical table :

Bonuses	frequency	Rel. Frequency	Cum. Frequency	Cum. Rel. Frequency
[750, 1250[	26	0.130	26	0.130
[1250, 1750[	71	0.355	97	0.485
[1750, 2250[	43	0.215	140	0.700
[2250, 2750[	26	0.130	166	0.830
[2750, 3250[	18	0.09	184	0.920
[3250, 3750[	10	0.05	194	0.970
[3750, 4250[	6	0.03	200	1.00

- Complete the statistical table above.
- What is the bonus value received by 50% of the employees.

Bonus value received by 50% of the employees ($n = 200$) represents the median $\tilde{X} \in [1750, 2250[$ and given by

$$\tilde{X} = L_1 + \left(\frac{\frac{n}{2} - C f_b}{f_m} \right) \times w = 1750 + \left(\frac{\frac{200}{2} - 97}{43} \right) \times (2250 - 1750) = \mathbf{1784.9}$$

- Give Q_1, Q_3 and inerquartile range

The first quartile Q_1 is located in the class $[1250, 1750[$ and the third quartle Q_3 is in $[2250, 2750[$ and both are given by

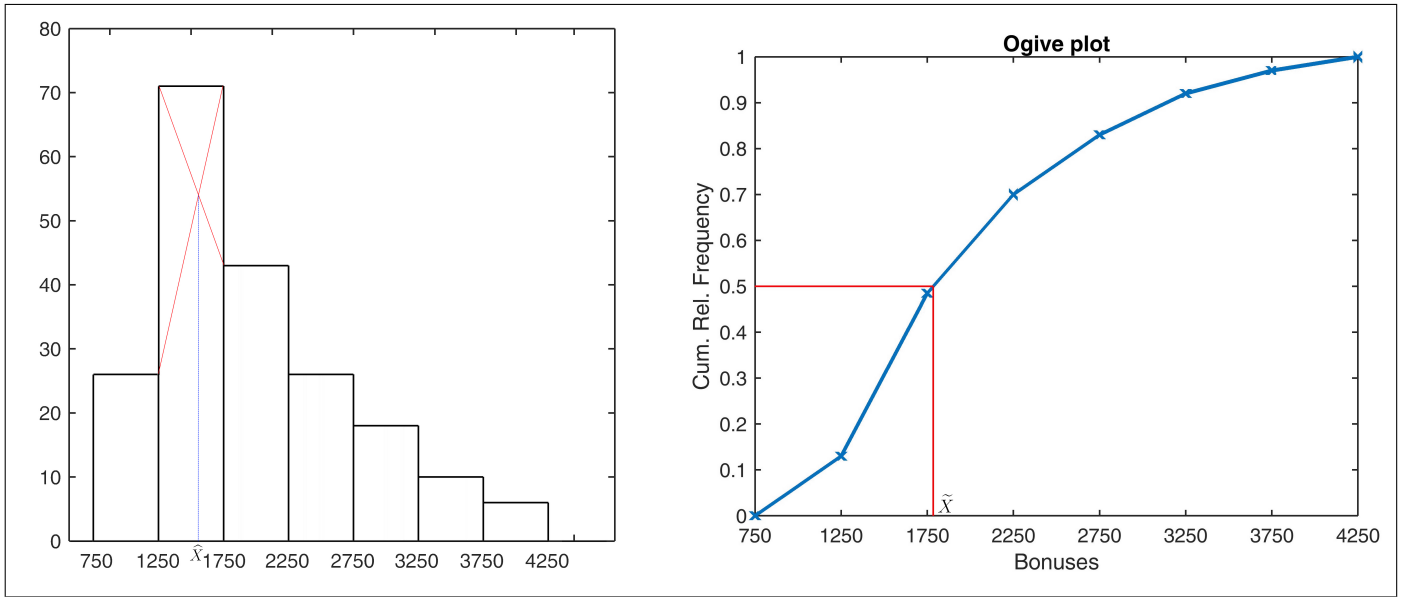
$$Q_1 = L_1 + \left(\frac{\frac{n}{4} - C f_{q_1}}{f_{q_1}} \right) \times w = 1250 + \left(\frac{\frac{200}{4} - 26}{71} \right) \times (1750 - 1250) = \mathbf{1419}$$

and

$$Q_3 = L_1 + \left(\frac{\frac{3n}{4} - C f_{q_3}}{f_{q_3}} \right) \times w = 2250 + \left(\frac{\frac{600}{4} - 140}{26} \right) \times (2750 - 2250) = \mathbf{2442.3}.$$

The inerquartile range is $IQR = Q_3 - Q_1 = 2442.3 - 1419 = \mathbf{1023.3}$.

- Plot the histogram and the Ogive plot of the above distribution then determine the mode and median of the series graphically



5. Give the modale class then calculate the mode value.

Clearly, The highest frequency is for the second class $[1250, 1750[$ to which the mode $\hat{X}$ belongs and it's given by

$$\hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w = 1250 + \left(\frac{71 - 26}{(71 - 26) + (71 - 43)} \right) \times (1750 - 1250) = \mathbf{1558.2}$$

6. Calculate the mean, variance, standard deviation and coefficient of variation.

Since the data are grouped, we need to compute the med point of each class and the formulas to be used are :

$$\mu = \bar{X} = \frac{\sum_{i=1}^7 f_i x_i}{\sum_{i=1}^7 f_i} = \frac{396500}{200} = \mathbf{1982.5}$$

$$\sigma^2 = Var(X) = \frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2 = \frac{8874643.75}{200} = \mathbf{44373.22}$$

and

$$\sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2} = \sqrt{44373.21875} = \mathbf{210.65}.$$

The coefficient of variation is given by

$$CV = \frac{\sigma}{\mu} \times 100 = CV = \frac{210.65}{1982.5} \times 100 = \mathbf{10.625\%}$$

7. Do you think the distribution of the performance bonuses is symmetric? justify.

No, the distribution of the performance bonuses is not symmetric characterized by the shape of the histogram, also by computing pearson's coefficient of skewness

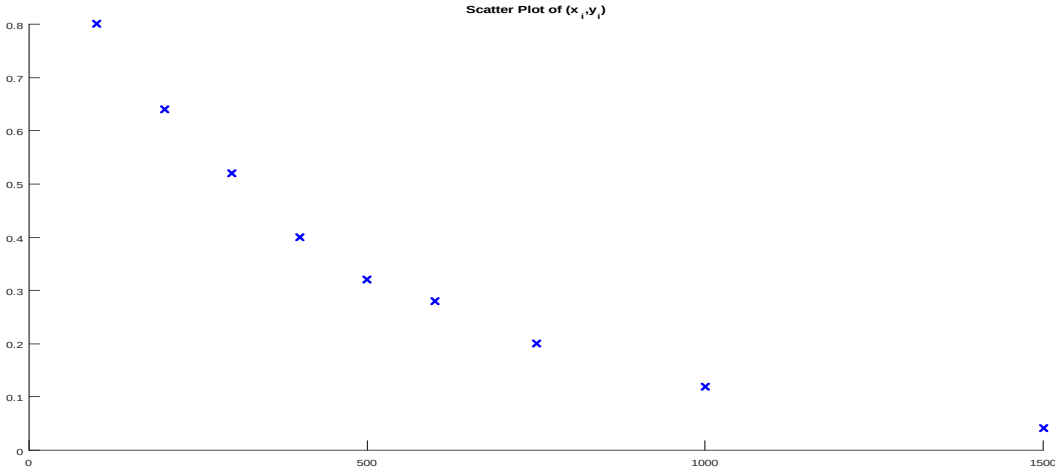
$$sk = \frac{3 \times (1982.5 - 1784.9)}{210.65} = \mathbf{2.8141}$$

we can observe that its value is close to 3 which let us conclude that the distribution of the performance bonuses is asymmetric

Exercise 2. The lifespan of a number of identical mechanical equipments was studied. In the following table, X represents the lifespan (expressed in hours) and Y is the percentage of equipment still in service in the date. For example, for $x_1 = 100$, 80% of the equipment remains in service since $y_1 = 0.8$.

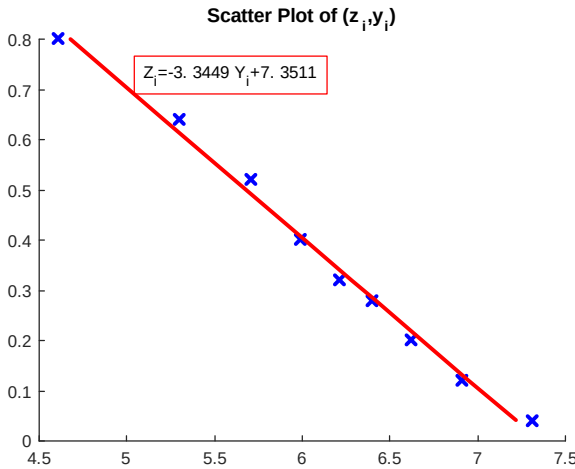
X	100	200	300	400	500	600	750	1000	1500
Y	0.8	0.64	0.52	0.4	0.32	0.28	0.2	0.12	0.041

1. Draw the scatter plot of the given data. Describe the relationship between X and Y .



Clearly, the relation between X and Y is not linear.

2. Set $Z = \ln(X)$, draw the scatter plot with coordinates (z_i, y_i) . Compare with the previous question and measure the correlation between them.



The relation between Z and Y is perfectly linear with a high negative correlation

$$\begin{aligned} \bar{Z} &= \frac{1}{9} \sum z_i = \frac{55.0513}{9} = 6.1168, \bar{Y} = \frac{1}{9} \sum y_i = \frac{3.3210}{9} = 0.369, \\ \sum (z_i - \bar{Z})^2 &= 5.5396, \sum (y_i - \bar{Y})^2 = 0.4914, \sum (z_i - \bar{Z})(y_i - \bar{Y}) = -1.6437. \end{aligned}$$

The above scatter plot suggested the presence of linear correlation, hence, we choose pearson's coefficient of correlation to measure it.

$$\rho(Z, Y) = \frac{Cov(Z, Y)}{\sigma_Z \cdot \sigma_Y} = \frac{\sum (z_i - \bar{Z})(y_i - \bar{Y})}{\sqrt{\sum (z_i - \bar{Z})^2} \sqrt{\sum (y_i - \bar{Y})^2}} = \frac{-1.6437}{\sqrt{5.5396} \times \sqrt{0.4914}} = -0.99624$$

which indicates the existence of a very strong negative linear correlation between the two variables.

3. Create a linear model to predict the values of Z then draw this line on the above scatter plot.

To predict the values of Z , we need first to fit the line

$$Z = aY + b + \varepsilon.$$

therefore, $\hat{a} = \frac{Cov(Y,Z)}{Var(Y)} = \frac{-1.6437}{0.4914} = -3.3449$ and $\hat{b} = \bar{Z} - \hat{a}\bar{Y} = 6.1168 - (-3.3449) \times 0.369 = 7.3511$ Thus

$$Z = -3.3449Y + 7.3511 + \varepsilon$$

4. Give the value of R^2 for the line of the 3rd question then briefly interpret its value.

We know that $Var(Z) = V_R(Z) + V_E(Z)$ which gives that

$$V_E(Z) = \frac{Cov(Y, Z)^2}{Var(Y)} = Var(Z)\rho(Y, Z)^2 = 0.61551 \times (-0.99624)^2 = 0.61089$$

$$V_R(Z) = Var(Z)(1 - \rho(Y, Z)^2) = 0.61551 \times (1 - 0.99624^2) = 4.6199 \times 10^{-3}.$$

The regression lines fit well the cloud of points in the above plot, to measure the goodness of fit, we need to compute the coefficient of determination

$$R^2 = \frac{V_E(Y)}{Var(Y)} = \frac{0.61089}{0.61551} = 0.99249$$

which is very close to 1.